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Governing agentic cities: platform cognition, power, and post-smart urbanism

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A B S T R A C T

The rise of artificial intelligence in urban governance signals a shift from sensor-based smart-city visions to post-smart urbanism, where machines not only sense but also learn, predict and act. This article interrogates whether AI necessitates new theoretical frameworks beyond smart-city paradigms by synthesising and comparing scholarship across nine interrelated dimensions—technology, governance, power, spatiality, autonomy, ethics, economy, culture and materiality. Building on this synthesis, we develop a conceptual toolkit for analysing post-smart cities and apply it in narrative case reviews of Hangzhou's City Brain initiative, Shanghai's AI development zone and the Praxis Network State. These cases illustrate how AI urbanism transforms governance, power relations, public space and infrastructural regimes: agentic AI systems reconfigure decision-making from human-in-the-loop analytics to algorithmic steering; platform urbanism concentrates data and compute power; and embodied AI technologies alter spatial codes, raising new concerns around surveillance, inequity and accountability. Our comparative analysis reveals that state–corporate alliances co-produce AI urbanism with divergent political economies and normative implications. Synthesising these insights, we propose a human-centred normative framework that aligns AI deployment with public value, emphasising inclusive governance, transparency, data sovereignty, environmental sustainability and participatory ethics. By clarifying key concepts, offering a comparative analytic framework and outlining policy recommendations, the study advances both theoretical and practical debates on how AI reconfigures urban life.

1. Introduction

The notion of the smart city emerged from early digital urbanism and gained prominence with the rise of the Internet of Things (IoT), in which sensors embedded in infrastructure produce streams of data to optimise energy grids, transportation and public services. In this paradigm, technology serves as a tool for planners: dashboards visualise urban metabolism and data assist human decision-makers. Nevertheless, the smart-city vision has been criticised for techno-centrism, surveillance, social inequity and its tendency to treat citizens as data points rather than active participants. Critics further argue that the smart city is not only techno-centric but also shaped by corporate narratives and 'solutionist' discourses, producing a technocratic approach that crowds out

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alternative visions (Sadowski & Bendor, 2019; Kitchin et al., 2019). Over the past few years, urban AI has come to the forefront. Large-scale AI systems such as city brains, autonomous vehicles, service robots and predictive algorithms now operate in urban domains. Recent work on platform urbanism shows that digital platforms extend across multiple cities and ecosystems, while webometric analyses reveal how the 'smart city' has become a global discourse with normative storylines (Caprotti et al., 2022; Joss et al., 2019). Scholars have thus begun to speak of a post-smart urbanism (defined in Section 1.1) paradigm, where artificial intelligences manifest in urban spaces, technologies and infrastructures and transcend the limitations of smart urbanism. Others warn that the convergence of AI and urbanism is controlled by surveillance-capitalist entities, threatening democratic values, while post-colonial studies reveal how smart-city projects deploy 'myths of speed' to impose technocratic futures (Makanadar, 2024; Datta, 2019). This new era raises profound questions about how cities are governed when machines not only sense but also reason and act.

This article investigates whether AI constitutes a new socio-technical logic that moves cities beyond the smart-city paradigm. We operationalise these nine dimensions in a cross-case design, using them as a comparative lens on Hangzhou, Shanghai and the Praxis Network State. While the promise of AI-driven efficiency and innovation has captured policy agendas, critical scholarship suggests the emergence of post-smart cities may amplify surveillance, exacerbate inequalities and erode democratic control. Existing research often overemphasises technological capabilities while neglecting the social, ethical and political dimensions of AI adoption. Consequently, there is a need to interrogate how AI reshapes urban governance, spatiality and power relations, and whether it necessitates new theoretical frameworks beyond those developed for smart cities. Such frameworks must integrate culture, urban metabolism and governance, while recognising the ecological footprint of AI—covering CO_2 emissions, data-centre energy consumption and e-waste—and embedding design cultures oriented toward sustainable, human-centred AI (Allam & Dhunny, 2019; Palmini & Cugurullo, 2024).

To address this problem, the article pursues four objectives. First, it compares smart and AI urbanism across nine analytical dimensions—technology, governance, power, spatiality, autonomy, ethics, economy, culture and materiality—to clarify what is distinctive about AI-enabled cities. Second, it develops a conceptual framework for the post-smart urbanism that integrates insights from urban innovation ecosystems and socio-technical systems theory. Third, it analyses three cases—Hangzhou's City Brain initiative, Shanghai's AI development zone and the Praxis Network State—to examine varied trajectories of AI urbanism, from state-led experiments to corporate-driven charter cities. Fourth, it identifies governance and ethical implications of AI adoption, and proposes policy recommendations for inclusive, transparent and sustainable urban AI. By bridging critical urban studies, technology ethics and innovation policy, the article contributes to emerging debates on post-smart urbanism and offers guidance for scholars and practitioners navigating the AI city.

1.1. Clarifying key concepts

Before proceeding, we clarify the key concepts used throughout this article and standardise terminology. We use post-smart urbanism as an overarching term for the socio-technical condition in which artificial-intelligence-enabled systems not only sense but also learn, predict, and act, thereby reconfiguring governance, power relations and urban space. This usage emphasises that the contemporary transformation of cities cannot be reduced to "smarter" versions of sensor-centric smart urbanism; instead, it marks the emergence of a new regime characterised by agentic autonomy and anticipatory governance. We distinguish this from AI urbanism, which refers more narrowly to the integration of artificial intelligence technologies into urban infrastructures and services. When referring to agentic cities, we foreground the presence of autonomous urban AI agents capable of goal-setting, planning and self-modification—an emphasis on agency that marks a shift from automation to autonomy (Tiwari, 2025). We use platform urbanism to highlight the socio-technical form in which digital platforms organise and mediate urban life by controlling data infrastructures, orchestrating services and shaping governance (Bratton, 2021; Caprotti et al., 2022). Throughout the article, post-smart urbanism functions as our master term; AI urbanism, agentic cities and platform urbanism are used to draw attention to specific technical or governance dimensions within this broader condition.

The article proceeds as follows. Section 2 reviews the evolution of smart cities, critiques their limitations, and surveys the literature on AI urbanism, platform and network formations, and urban innovation ecosystems; it then synthesises these debates into a nine-dimensional conceptual framework for post-smart urbanism and presents the proposed theoretical model. Section 3 outlines the research design and methodology, including the literature synthesis, data sources and analytical approach, and introduces the three case studies—Hangzhou's City Brain initiative, Shanghai's AI development zone and the Praxis Network State. Section 4 presents a cross-case synthesis and discussion of findings, focusing on governance challenges, socio-technical synergies, ethical concerns, and political-economic dynamics. Section 5 concludes by summarising the article's contributions, limitations, and implications for theory, policy, and future research.

2. Literature review and theoretical foundations

To orient readers, this section lays the groundwork for our conceptual framework by moving from the evolution of smart cities toward post-smart urbanism (Section 2.1), differentiating AI urbanism from smart urbanism (Section 2.2), examining the role of innovation ecosystems and governance (Section 2.3), exploring the intersection of AI and digital authoritarianism (Section 2.4), considering the specific trajectory of urban AI in China (Section 2.5), and addressing the ethical, social and environmental concerns associated with urban AI (Section 2.6). Section 2.7 then synthesises these insights into a conceptual framework anchored in nine analytical dimensions.

2.1. From smart to post-smart urbanism

Smart cities emerged as a response to rapid urbanisation and the diffusion of digital technologies. They rely on extensive sensor networks and the Internet of Things (IoT) to collect real-time data about urban systems and feed that information to dashboards that aid human decision-makers (Cugurullo et al., 2024). In this paradigm, technologies such as smart grids, automated transit systems and ubiquitous monitoring aim to optimise energy use, mobility and service delivery while maintaining human control over decisions. Corporate narratives crafted by major technology firms—frequently presenting turnkey, data-driven 'solutions' to urban crises—have reinforced a solutionist logic that privileges technocratic fixes and sidelines alternative, more democratic or situated perspectives (Sadowski & Bendor, 2019; Kitchin et al., 2019). Despite promises of efficiency, smart-city initiatives have been critiqued for reinforcing socio-spatial inequalities and reproducing neoliberal governance logics. Scholars argue that sensor-driven approaches often reduce citizens to data points and prioritise market solutions, leading to surveillance, exclusion and a lack of democratic participation (Bai et al., 2025). Postcolonial scholarship further shows that smart-city discourses can reproduce colonial narratives and 'moral states,' mobilising myths of speed and inevitability that naturalise technocratic futures and obscure questions of power, history and temporality (Datta, 2019). The uneven deployment of smart technologies thus exacerbates existing disparities and raises concerns about privacy and accountability. As Datta and Odendaal (2019) note, this 'banality of power'—the normalisation of control through seemingly mundane technical routines—allows smart-city projects to entrench authority while appearing neutral.

Analyses of big data and AI in cities also caution against narrow technocentric framings, calling instead for integrative frameworks that braid sustainability, culture and livability into the design and governance of urban technologies (Allam & Dhunny, 2019). Such approaches foreground public value, institutional capacity and environmental limits alongside digital innovation. In recent years, researchers have proposed moving beyond the smart city toward post-smart urbanism, where artificial intelligences (AIs) operate within urban spaces and infrastructures. This paradigm envisions city brains, robots and autonomous systems that not only sense but also learn, predict and act, thereby transcending the technological, governance and material limitations of smart urbanism (Bai et al., 2025). By embedding AI in everyday objects and public spaces, post-smart cities shift focus from passive data collection to active decision-making and anticipatory governance, setting the stage for new configurations of urban power, agency and space. Accordingly, Caprotti et al. (2024) argue that urban-AI technologies exercise distinct social and material power, calling for research agendas in urban studies that interrogate these effects and their consequences for urban life.

2.2. AI urbanism versus smart urbanism

AI urbanism refers to the integration of artificial intelligence into the fabric of cities, producing autonomous systems that influence how cities function and are governed. Unlike smart urbanism, which relies on IoT and human oversight, AI urbanism involves the Artificial Intelligence of Things (AIoT)—technologies that sense and act autonomously (Cugurullo et al., 2024). Table 1 summarises key differences across nine categories that are later operationalised as analytical dimensions in Section 2.7.2. A comprehensive analysis of smart urbanism must attend to layered urbanism, governance configurations and citizen engagement; as Verrest and Pfeffer (2019) argue, this perspective extends and deepens the nine-dimensional framework advanced here by foregrounding how these layers interact. Crucially, AI urbanism marks a shift from automation to autonomy: urban AI (UAI) technologies such as self-driving cars and service robots begin to take management out of human hands, introducing autonomous agents into planning and governance (Cugurullo, 2020). Within the "Autonomy" dimension, agentic AI systems can set goals, plan, and modify behaviour in response to changing environments, raising novel governance and ethical challenges (Tiwari, 2025). Materially, AI urbanism introduces embodied systems—robots, drones, autonomous vehicles—that occupy, reconfigure and contend for public space, intensifying questions of socio-spatial sorting and responsible urban innovation (Palmini & Cugurullo, 2023; Macrorie et al., 2021). Relatedly, platform urbanism remains fragile and speculative in practice: 'glitchy' breakdowns and future-casting imaginaries shape how systems are built and experienced (Leszczynski, 2016, 2020). At the limit, smart-city logics have been critiqued as forms of 'cybernetic urbanism' tending toward a 'society of control' (Krivý, 2018), a concern that intensifies as decision prerogatives migrate to autonomous agents. These distinctions underscore that AI urbanism is not merely an extension of smart urbanism but a qualitatively different socio-technical regime, with implications for power, economy and culture alongside technology and governance (Caprotti et al., 2022; Sanchez et al., 2025; Wood, 2025).

Table 1
Smart urbanism versus AI urbanism.

Category	Smart urbanism (keywords)	AI urbanism (keywords)
Technology	IoT sensors; big-data dashboards; edge collection	AIoT; on-device/edge inference; autonomous actuators
Governance	Human-in-the-loop analytics inform policy	Algorithmic policy steering; "city brain" orchestration; automated enforcement
Power	Shared public–private control; municipal oversight	Platform/data concentration; state–tech data sovereignty; vendor lock-in
Spatiality	Embedded, largely invisible infrastructure	Visible robots/drones/AVs; reprogrammed streets/curbs; geofenced zones
Autonomy	Automation (rule-based, scripted repetition)	Agentic autonomy (goal-setting, planning, self-adaptation; machine decision-making)
Ethics	Privacy/compliance framing; consent by design	Bias/fairness/explainability risks; accountability gaps; pervasive surveillance
Economy	PPPs; procurement-led "solutions"; vendor ecosystems	Platform capitalism; data monetisation; charter-city/Network-State models
Culture	Impersonal devices; civic-tech rhetoric	Anthropomorphised agents (named robots/voice assistants); everyday AI culture
Materiality	Lightweight sensors/microchips; server reliance	Heavy robotics/AV fleets; control rooms; data centres; charging/robot lanes

2.3. Urban innovation ecosystems and governance

The concept of urban innovation ecosystems frames cities as networks of public agencies, private firms and citizens co-creating technological and social solutions. Complementing this ecosystem view, the notion of urban operating systems shows how computational logics and platform architectures coordinate urban functions and structure decision rights, thereby shaping governance itself (Luque-Ayala & Marvin, 2020). Effective ecosystems depend on robust urban fabric, knowledge exchange and civic engagement, with municipalities increasingly acting as ecosystem stewards that orchestrate collaborations to enhance public value. While much smart-city scholarship foregrounds technical infrastructures, recent work stresses that durable innovation hinges on social capacity—participatory decision-making, institutional learning and mechanisms that render algorithmic systems publicly accountable. This shift demands updated accountability architectures for algorithmic governance—including transparency, contestability and clearly allocated institutional responsibility (Katzenbach & Ulbricht, 2019). At the same time, contemporary urban AI should be analysed through the lens of assemblages and imaginaries that mobilise policies, publics and design cultures (Lazzeroni, 2023). Systematic reviews of AI adoption in urban planning emphasise the need to secure digital rights, strengthen public accountability and embed social oversight, noting that AI is evolving from analytical support toward co-decision-making in governance processes (Lartey & Law, 2025). Achieving socio-technical synergy therefore requires governance arrangements that align technical performance with civic legitimacy, distributional equity and environmental limits. Evidence from AI-driven innovation in smart-city governance further indicates that deployments should be oriented to human-centric and sustainable outcomes, coupling performance gains with ethical safeguards and social value creation (Bosco et al., 2024). Given that Urban AI is still an emergent field, participatory and reflexive approaches are essential to navigate uncertainty and embed public values across the lifecycle (Luusua et al., 2023).

AI localism has emerged as a practical response to this challenge: cities are often moving faster than national governments to craft rules for AI, focusing on normative principles, public engagement, transparency requirements, impact assessments and responsible procurement (Marcucci et al., 2022). Local charters, procurement clauses and audit protocols operationalise these commitments—-translating abstract ethical aspirations into enforceable obligations on vendors and agencies, and creating procedural venues (e.g., public registers, complaint channels, citizen assemblies) through which residents can contest and shape AI deployments. Finally, human-centric design is a throughline for ecosystem governance. Evidence from European and municipal pilots indicates that AI must be configured to balance efficiency gains with sustainability targets and equitable access to services, avoiding narrow optimisation that externalises social or environmental costs (Ejjami, 2024). Design cultures oriented toward sustainable urban AI—including attention to material footprints, cultural context and everyday practices—provide a pathway to embed human values across the lifecycle of urban AI systems, from problem framing and data governance to deployment and decommissioning (Palmini & Cugurullo, 2024).

2.4. The network state and digital authoritarianism

Clear connections to Bratton (2019) on issues of governance in the context of planetary scale computation

Tech entrepreneurs have proposed the Network State: privately governed, AI-driven city-projects that operate beyond conventional state regulation (Wood, 2025). Such proposals imagine blockchain-, cryptocurrency- and AI-powered settlements functioning as quasi-independent polities. The Praxis Network State represents an extreme post-smart vision: corporate-led communities characterised by ubiquitous surveillance, exclusionary security regimes and the prioritisation of property over democratic rights—market freedoms wrapped in an aesthetic politics of escape for a technological elite. Critics contend that this libertarian rhetoric can mask authoritarian, racist and eugenicist tendencies, illustrating how post-smart urbanism may enable techno-authoritarianism if left unchecked (Wood, 2025). This trajectory resonates with Krivý's (2018) critique of cybernetic urbanism as a 'society of control,' wherein feedback-driven optimisation governs populations while sidelining democratic deliberation. These tendencies are reinforced by surveillance-capitalist business models that extract behavioural data, commodify urban life and externalise public risk, thereby threatening democratic accountability and civic freedoms. Concentrated data power and opaque algorithmic governance can rewire urban sovereignty around platform interests, deepening inequalities in participation and rights (Makanadar, 2024). In response, decentralised and community-centred approaches offer partial counterweights. Citizen-driven IoT initiatives demonstrate how bottom-up innovation and community co-design can reframe sensing and governance around local priorities, strengthening trust and accountability (Kummitha & Crutzen, 2019). Emerging experiments in AI for social innovation leverage distributed ledgers, data cooperatives and community trusts to enhance data sovereignty, widen participation and align AI deployments with local public value. While not a panacea, such models illustrate pathways to redistribute power and embed accountability within post-smart governance architectures (Calzada, 2024). This debate clarifies how post-smart trajectories can align governance, autonomy, and power toward exclusionary or authoritarian outcomes, motivating the framework's attention to political economy and accountability.

participatory design

2.5. Urban AI in China: Navigating social control and hyper-capitalist development

Building on Marvin et al.'s (2022) framing of Chinese urban AI experiments as oscillating between social control and hyper-capitalist development, this section examines how Shanghai's AI development zone and Hangzhou's City Brain initiative illustrate these tensions and add further complexity, especially when viewed through spatiality and materiality. Comparative studies reveal both similarities and divergences with smart-city discourse, producing a landscape that cannot be neatly reduced to either social control or market efficiency (Marvin et al., 2022). Several drivers underpin China's urban AI push: a national AI strategy that aims to make China a global leader in AI innovation; rapid urban growth and the need to manage congestion and pollution; and the integration of platform economies and ubiquitous surveillance infrastructure. Major technology companies such as Alibaba, Tencent and Baidu collaborate with municipal governments, illustrating the convergence of corporate and state interests in developing AI solutions. Hangzhou's City

Brain exemplifies platform urbanism in the Chinese context: it is co-produced through public–private partnerships (notably with Alibaba) and territorialised through incremental experiments that embed AI within municipal governance routines—from traffic to health and city operations—illustrating how platform logics are institutionalised in place (Marvin et al., 2022). Technically, City Brain also stands as an early practice of large-scale AI deployment in a real urban environment, integrating citywide data streams with machine-learning pipelines to orchestrate services at scale (Zhang et al., 2019). Beyond mobility and city operations, AI megaprojects in China's healthcare services—developed through state–corporate partnerships—further demonstrate the cross-sectoral reach of AI urbanism and its consolidation through national-scale deployments (Guo & Cugurullo, 2025). Alongside platformised governance, urban robotics experimentation has been a salient strand of China's post-smart trajectory, echoing global testbeds in cities such as San Francisco, Tokyo and Dubai (While et al., 2021). Trials involving delivery robots, autonomous shuttles and drone logistics signal a material shift in public space management and regulatory attention, complementing data-centric "city brain" architectures with embodied AI systems that directly interact with—and reorganise—urban space (While et al., 2021; Zhang et al., 2019). The Chinese case illustrates how governance, power, and economy co-produce urban AI at scale, while also foregrounding spatiality and materiality through city-brain infrastructures and robotics testbeds.

2.6. *Ethical, social and environmental concerns*

As AI assumes decision-making roles in urban systems, ethics comes to the fore: planners and municipalities must confront risks of bias, opacity and weak accountability chains, alongside heightened privacy exposure in data-intensive services. In AI-infused IoT and smart-city stacks, legal and sectoral analyses foreground data-protection, liability and human-rights risks that must be addressed in system design and governance (Rawindaran, 2023; Shafik, 2024). Recent guidance in planning scholarship recommends inclusive and representative datasets, public engagement throughout the AI lifecycle, and continuous post-deployment monitoring and auditing to mitigate discriminatory outcomes and ensure due process (Sanchez et al., 2025). A recent systematic review likewise shows AI is already reshaping urban design and planning, underscoring the need for interdisciplinary oversight (He & Chen, 2024). In parallel, work on algorithmic governmentality warns that smart-city rationalities can hard-code unequal geographies of service and policing, reproducing spatial injustice under the guise of optimisation (Rodrigues, 2016). At the modelling layer, machine-learning applications for urban land-use planning are expanding (Chaturvedi & de Vries, 2021) but demand accountability and transparency. Cities should document algorithmic objectives, datasets and validation protocols, publish impact assessments and maintain audit logs, while limiting overbroad trade-secret claims that block scrutiny (Baykurt, 2022; Brauneis & Goodman, 2018). At the interface of materiality, spatiality and autonomy, robotics and automation extend human agency into fleets of embodied systems (delivery robots, drones, autonomous shuttles) that reorganise curb space, logistics and micromobility. These trials surface issues of socio-spatial sorting, accessibility and the governance of mixed human–robot environments (Macrorie et al., 2021). Autonomous vehicles add broader societal externalities—reconfiguring labour markets, travel behaviour and risk allocation—while potentially amplifying inequalities if benefits and burdens are unevenly distributed (Bissell et al., 2020).

though, as mendonca (2022) illustrates, diversity and representation in data does not always result in equitable design - that has to occur intentionally on the part of planners and other urban designers

Innovation and sustainability frameworks argue that AI should be explicitly tied to urban innovation goals and ethical principles—fairness, human well-being and ecological limits—within city governance (Ortega-Fernández et al., 2020; Pastor-Escuredo et al., 2022). The environmental footprint of urban AI is non-trivial: training and operating models, powering data centres and manufacturing robotic hardware entail CO_2 emissions, water and energy demands, and growing streams of e-waste. A 'Green AI' agenda therefore calls for energy-efficient, equitable and environmentally sustainable systems (Yigitcanlar et al., 2021). A sustainable AI design culture therefore calls for lifecycle assessments, repairability, and circular practices to align innovation with climate and resource goals (Palmini & Cugurullo, 2024). Finally, temporal governance is shifting as generative AI furnishes predictive narratives that act as "oracles," enabling anticipatory governance wherein projections of the future shape present decisions. While potentially valuable for foresight, over-reliance on such oracular outputs risks locking-in, narrowing policy imagination and displacing human judgment in contested domains (Cugurullo & Xu, 2025). Culturally, the everyday presence of anthropomorphised agents (named robots, voice assistants) can foster convenience and trust but may also blur accountability and normalize pervasive sensing (Cugurullo et al., 2024). Taken together, these concerns span the article's nine dimensions—technology, governance, power, spatiality, autonomy, ethics, economy, culture, and materiality—and underscore the need for explainable, accountable AI, environmental due diligence, and procedural venues through which affected publics can contest and recalibrate AI-assisted urban decisions. Taken together, the preceding subsections underscore the need for an integrative framework that accounts for technological, governance, political-economic and ethical dynamics; the next subsection therefore develops such a framework.

2.7. *Conceptual framework for the post-smart urbanism*

2.7.1. *Defining post-smart urbanism*

Synthesising the foregoing literature, post-smart urbanism refers to an urban socio-technical condition in which AI systems not only sense but also learn, predict, and (in some domains) autonomously act upon urban processes. In this regime, AI is integrated into the core of governance and infrastructure, enabling anticipatory and self-adjusting management of mobility, energy, safety and public services. When we refer to "post-smart cities," we mean concrete empirical instantiations of post-smart urbanism in specific places and governance settings. Strategic frameworks for urban AI further argue that the field must integrate technology systems, application scenarios, education and governance—and be approached through interdisciplinary models capable of addressing complexity and hybrid human–machine systems (Zhu & Liu, 2025). This reconfiguration of urban life transcends IoT-centred optimisation, producing new forms of human–AI co-governance, novel political economies—including platform capitalism and network states—and expanded ethical challenges encompassing autonomy, privacy and equity (Bai et al., 2025; Wood, 2025; Marvin et al., 2022). Building on agentic AI theories, post-smart cities increasingly feature autonomous urban agents whose behaviours can adapt and self-modify, which in turn demands participatory and transparent governance arrangements to sustain legitimacy (Tiwari, 2025).

2.7.2. Analytical dimensions

To analyse post-smart urbanism, we synthesise the literature into nine interconnected dimensions. Table 1 summarises how smart and AI urbanism diverge across these categories. Below, we elaborate on each dimension, drawing on key contributions to highlight how these debates inform our conceptual model.

Technology – Post-smart urbanism extends beyond IoT sensors and big-data dashboards to AIoT systems that sense, learn and act. These systems embed AI capabilities in edge devices and autonomous actuators, enabling real-time inference and intervention (Zhu & Liu, 2025; Cugurullo et al., 2024).

Governance – AI alters decision-making by shifting from human-in-the-loop analytics to algorithmic policy steering. "City Brain" architectures orchestrate municipal operations, while predictive systems blur the line between recommendation and automation. Scholars emphasise updated accountability architectures, transparent procurement, and participatory oversight (Katzenbach & Ulbricht, 2019; Lartey & Law, 2025; Marcucci et al., 2022).

Power – Control over AI infrastructure and data confers new forms of infrastructural power, raising concerns about platform concentration and state–corporate alliances. Platform capitalism and network-state ventures illustrate how AI urbanism can consolidate economic and decision-making power, while citizen-driven models and data cooperatives offer counterweights (Barns, 2020; Calzada, 2024).

Spatiality – AI reconfigures the spatial organisation of cities by introducing visible robots, drones and autonomous vehicles into public space. These embodied systems reprogram streets, sidewalks and curbs, creating geofenced zones and new logistical corridors; they also intensify socio-spatial sorting and call for new design codes to govern human–robot interactions (Palmini & Cugurullo, 2023; Macrorie et al., 2021).

Autonomy – Post-smart urbanism is defined by a shift from automation to agentic autonomy. Agentic AI systems set goals, plan and adapt behaviour in dynamic environments, introducing a machine-in-the-loop governance model and raising questions about meaningful human control and accountability (Cugurullo, 2020; Tiwari, 2025).

Ethics – The ethical dimension encompasses fairness, bias, transparency, accountability and privacy. Scholars underscore inclusive data practices, algorithmic explainability, impact assessments and auditing to mitigate discriminatory outcomes and safeguard rights (Sanchez et al., 2025; Baykurt, 2022). The environmental footprint of AI also demands ethical consideration and life-cycle governance

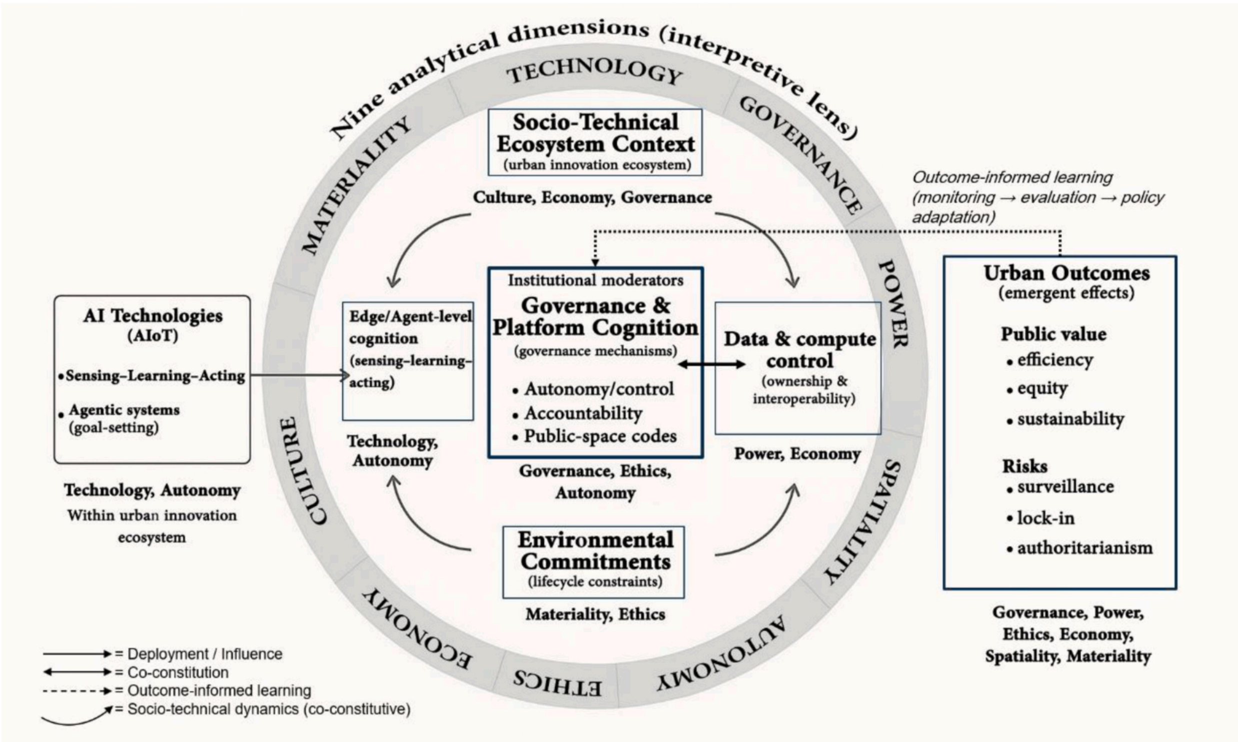

Fig. 1. Conceptual framework for post-smart urbanism. The outer ring presents the nine analytical dimensions as an interpretive lens. Component labels indicate the primary dimensions each cluster speaks to. Arrow styles indicate deployment/influence (solid), co-constitution (double-headed), outcome-informed learning (dashed), and broader socio-technical dynamics (curved).

(Palmini & Cugurullo, 2024).

Economy – AI urbanism underpins new economic models such as platform capitalism, data monetisation and charter-city ventures. These models reconfigure value chains by capturing and commodifying urban data, raising questions about public value capture and redistribution (Caprotti et al., 2022; Wood, 2025).

Culture – Everyday interactions with anthropomorphised agents—named robots, voice assistants and chatbots—cultivate an AI culture that normalises machine agents as social actors. This dimension also encompasses imaginaries and narrative framings that shape public understanding and acceptance of urban AI (Joss et al., 2019; Lazzeroni & Romano, 2025).

Materiality – Post-smart urbanism is materially grounded in heavy robotics, AV fleets, control rooms, data centres and edge devices. These hardware infrastructures entail significant energy consumption, CO_2 emissions and e-waste, prompting calls for Green AI and circular design practices (Yigitcanlar et al., 2021; Palmini & Cugurullo, 2024).

These paragraphs synthesise the literature across the nine dimensions. For transparency, Supplementary Appendix A (Table S1) provides an index mapping core sources to each analytical dimension. These dimensions are interdependent rather than additive: advances in AIoT and agentic autonomy reconfigure governance arrangements and redistribute power, which in turn reshape spatial organisation and material infrastructures (e.g., streets, curbs, data centres). The consequences are distributive and ecological—so ethical commitments and accountability mechanisms function as constraints on how technological capability translates into urban outcomes.

2.7.3. *Proposed theoretical model*

Building on the nine analytical dimensions synthesised in Section 2.7.2, Fig. 1 operationalises them as a socio-technical model linking AI capabilities, governance arrangements, and urban outcomes. The outer ring presents the nine dimensions as an interpretive lens. Within the model, AI Technologies (AIoT) correspond to Technology, while edge/agent-level cognition captures the shift toward Autonomy through sensing–learning–acting and goal-setting systems. Governance & Platform Cognition instantiates Governance and specifies institutional mechanisms through which AI is steered (e.g., autonomy/control, accountability, and public-space codes), linking governance to Ethics and Autonomy. Data & compute control foregrounds Power and Economy by emphasising ownership, interoperability, and infrastructural control. The Socio-Technical Ecosystem Context situates these dynamics within broader cultural–economic–institutional conditions, while Environmental Commitments introduce lifecycle constraints connecting Materiality and Ethics. Arrow styles distinguish: (1) deployment/influence (solid), (2) co-constitution (double-headed), (3) outcome-informed learning (dashed; monitoring → evaluation → policy adaptation), and (4) broader socio-technical dynamics (curved arrows) that capture iterative co-evolution across the ecosystem. Read together, the model clarifies how the nine dimensions operate as an interconnected explanatory framework rather than a checklist.

3. Methodology

3.1. *Scope and purpose*

This study employs a mixed-method qualitative approach that combines critical literature synthesis, comparative case studies and normative analysis. The literature synthesis reviews scholarship on smart cities, AI urbanism, innovation ecosystems and digital authoritarianism to build a theoretical foundation and identify analytical dimensions. Three case studies—Hangzhou's City Brain initiative, Shanghai's AI development zone and the Praxis Network State—are selected because they are prominent examples in post-smart city discourse, represent diverse political economies (state-led public–private partnership in China versus corporate-led charter city proposals) and offer accessible published data. The case selection also deliberately draws on large-scale AI implementations such as the City Brain model, which has been operationalised at urban scale in Hangzhou, and on insights from urban robotics experimentation programmes in global testbeds (Zhang et al., 2019; While et al., 2021). Normative analysis is woven throughout to assess the ethical and governance implications of AI adoption. We conducted a structured literature scan and purposive case selection. Inclusion criteria for literature were: (i) English-language peer-reviewed publications between 2016 and 2025, supplemented by grey literature (policy documents, official white papers, standards, municipal ordinances) when they contained original data, explicit methods, or official technical specifications; (ii) substantive relevance to smart cities/AI urbanism, urban robotics/AVs, platform governance, or AI localism; and (iii) clear linkage to one or more of the nine analytical dimensions. Exclusions included opinion pieces lacking empirical or methodological substance, non-urban AI, and sources outside the window unless canonical to the argument. Case inclusion criteria were: (i) recognised exemplars in post-smart discourse; (ii) availability of multi-source documentation enabling triangulation; (iii) diversity of political economies and governance arrangements; and (iv) evidence of AI systems moving from sensing/analytics to acting/agentic operation. Given the conceptual nature of this study, the case studies are used to illustrate how the nine dimensions manifest across diverse governance contexts rather than to generate new empirical findings. Our analysis therefore draws on secondary sources and employs cross-case comparative logic to identify patterns. The cases were selected purposively to capture variation in governance and political economy: Hangzhou exemplifies a state-led public–private partnership; Shanghai represents a nationally supported municipal testbed; and the Praxis Network State serves as a theoretical limit case illustrating normative extremes of privately governed AI urbanism. By juxtaposing these cases, we trace dynamics of power, autonomy and governance across divergent models.

3.2. Data sources and collection

The study draws on secondary sources including peer-reviewed literature, policy documents, white papers, media reports and data from AI city initiatives. The literature review window was 2016–2025, with grey literature included only when it provided primary data, official technical detail, or clearly specified methods. For the Hangzhou City Brain case, we rely on the detailed case study by Bai et al. (2025), which provides interviews, archival documents and qualitative analysis of governance mechanisms. We also consult technical and practice-oriented accounts of City Brain deployments to triangulate operational details (Zhang et al., 2019). For the Shanghai AI development zone, information comes from Marvin et al. (2022), which synthesises policy documents, empirical observations and expert interviews. The Praxis Network State case is analysed through the critical commentary by Wood (2025) and supplementary reports on network-state and charter city experiments. Additional secondary data, such as news articles and corporate materials, are consulted to contextualise NEOM, Masdar City and Toronto Quayside. To complement these sources, we review the AI-localism literature that documents how cities develop AI policies, principles, engagement practices and responsible procurement often ahead of national regulation (Marcucci et al., 2022). We further incorporate studies on generative-AI adoption in city ecosystems and anticipatory governance to capture emergent practices and risks (Chandy, 2025; Cugurullo & Xu, 2025). Finally, we draw on urban-robotics scholarship to understand how automation pilots and testbeds shape socio-spatial experimentation and governance (While et al., 2021).

3.3. Analytical methods

Interpretation of each case is guided by the analytical dimensions outlined in Section 2.7.2 (technology, governance, power, spatiality, autonomy, ethics, economy, culture and materiality). For each dimension, we examine how AI technologies are deployed, who controls them, how they shape urban space and social relations, and what ethical challenges arise. Cross-case comparison identifies common themes, such as the shift from IoT to AIoT and the tension between efficiency and surveillance, as well as divergences in political economy and governance. Normative analysis evaluates whether AI initiatives enhance public welfare or entrench authoritarian or hyper-capitalist tendencies. Because evidence is derived from secondary sources rather than fieldwork, it may reflect publication and reporting biases; we triangulate across source types and treat inferences as provisional. The case analyses presented in Section 3.4 are therefore structured as narrative reviews of existing empirical studies rather than presentations of original fieldwork data. This methodological choice positions the article as a conceptual and integrative contribution: by synthesising empirical findings across diverse contexts, we derive cross-case insights that refine our theoretical framework and normative recommendations.

3.4. Case studies

3.4.1. Hangzhou city brain initiative

The Hangzhou City Brain (HCBI) began in 2016 as a technology-driven effort to alleviate traffic congestion by integrating big-data analytics and AI into urban transportation. Stage 1 focused on traffic management; by October 2017, version 1.0 of the system was released. Stage 2 shifted toward context-driven urban governance: by March 2018, HCBI expanded from transportation to broader urban management, adding services such as medical treatment, tourism and community development. Stage 3 emphasised ecosystem-driven innovation: in December 2019, Hangzhou released version 3.0 of its City Brain architecture and new digital public services. During the COVID-19 pandemic, the HCBI introduced a widely used health-code system that integrated vaccination, testing and travel information. Governance mechanisms identified in Bai et al. (2025) include the integration of public agencies, private companies (notably Alibaba) and citizens within an urban innovation ecosystem, enabling socio-technical synergy. Semi-structured interviews revealed that stakeholders collaborated through formal and informal networks, supported by municipal leadership and digital platforms. Outcomes of HCBI include improved traffic flow, smarter healthcare triage and educational services, and the "digital cockpit" that allows officials to oversee city operations in real time. To provide a deeper empirical picture, we draw on secondary sources to map the iterative development of the City Brain. During its first phase (2016–2017), the system integrated CCTV feeds, GPS data and weather information to optimise traffic signals, reportedly reducing congestion by 15% and cutting average travel times by 8% (Cugurullo, 2021). In Stage 2 (2017–2019) the platform expanded beyond traffic to coordinate police dispatch, environmental monitoring, hospital triage and tourism services. Stage 3 (2019–present) introduced open APIs and supported third-party applications, while a 'digital cockpit' provided real-time dashboards to municipal leadership. The COVID-19 pandemic prompted a further innovation: a health-code system based on City Brain architecture that integrated vaccination records, testing results and travel histories into a single QR code widely used for access control. Stakeholders interviewed by Bai et al. (2025) attributed the initiative's effectiveness to municipal coordination capacity, Alibaba's cloud platform and citizen feedback via super-apps. These developments underscore how AI urbanism in Hangzhou is co-produced by public and corporate actors and yields tangible impacts on mobility, public health and service delivery. Nevertheless, concerns persist about data governance, citizen participation and equity; reliance on a single tech conglomerate concentrates power and raises privacy issues. Practice-level accounts inform the operational profile of HCBI and its platform architecture (Zhang et al., 2019).

3.4.2. Shanghai AI development zone

Shanghai was designated a National Pilot Zone for New-Generation AI and serves as a test bed for AI technologies. According to Marvin et al. (2022), the city partners with leading tech firms to build data infrastructure, including cloud platforms, sensor networks and AI research hubs. Experiments encompass traffic-management algorithms, logistics optimisation and large-scale surveillance systems. Municipal authorities collaborate with companies through joint ventures and innovation centres; the dual aim is to spur economic growth and demonstrate AI's governance potential. This blended mode of AI/human decision-making is evident in systems that suggest interventions (e.g., rerouting traffic) while leaving final decisions to city officials. However, as AI capabilities expand, the boundary between recommendation and automatic action blurs, heightening tensions between innovation and social control. Related urban-robotics experimentation provides additional context for how cities trial automation and integrate lessons into governance (While et al., 2021).

3.4.3. Praxis network state and charter city experiments

The Praxis Network State (PNS) proposes an extreme form of post-smart urbanism: a privately governed city built on blockchain and AI platforms, funded by venture capital and designed to operate outside conventional state regulation (Wood, 2025). Unlike the Hangzhou and Shanghai cases, which draw on empirical observations, the PNS example functions as a theoretical limit case. We include it to illustrate how AI urbanism might evolve under corporate sovereignty and to surface normative issues around exclusion, rights and algorithmic rule; the analysis therefore synthesises critical commentary rather than summarising primary data. PNS imagines a community of entrepreneurs and technophiles who would secede from existing polities, adopt cryptocurrency as the medium of exchange and rely on algorithms for governance. Wood (2025) argues that the ideology behind PNS is libertarian–authoritarian: while advocating market freedoms, it envisages exclusionary citizenship, ubiquitous surveillance and the prioritisation of property rights over democratic participation. The network state movement, including charter cities like Prospera in Honduras, has been criticised for masking racism, technofascism and eugenics behind the rhetoric of innovation. These projects demonstrate how AI and blockchain can be mobilised to create gated urban enclaves for a global elite, raising questions about equity and rights.

3.4.4. Comparative analysis and additional examples

Beyond China and the Praxis proposals, other AI-driven urban projects illustrate the diversity of post-smart trajectories. NEOM in Saudi Arabia markets itself as a fully automated, AI-managed region that will reinvent living and working; Masdar City in Abu Dhabi integrates renewable energy and smart technologies; Toronto Quayside (the Sidewalk Labs project) sought to build a data-driven neighbourhood before public backlash halted it. These examples reveal patterns of corporate–state collaboration, heavy reliance on proprietary technology and significant public resistance due to privacy and governance concerns. Our cross-case reading is complemented by AI-localism analyses that document city-level governance innovations and by accounts of generative-AI deployments in civic and street-level contexts, which inform the anticipatory and socio-technical dynamics of emerging AI uses (Marcucci et al., 2022; Chandy, 2025; Cugurullo & Xu, 2025). These comparisons (Table 2) underscore that the post-smart urbanism is not monolithic but varies across contexts, driven by different political economies and subject to contestation.

3.5. Case selection rationale and methodological reflections

The selection of Hangzhou, Shanghai and the Praxis Network State reflects our intention to capture the diversity of AI urbanism rather than to provide a comprehensive empirical survey. We purposively selected cases that are widely referenced in the literature and represent distinct political economies and governance configurations. Hangzhou's City Brain is a state-led public–private initiative that is operational at scale; Shanghai's AI development zone is a municipal testbed embedded in national strategy; and the Praxis Network State is included as a theoretical limit case, illustrating speculative and corporate-led trajectories that inform debates on AI urbanism. The Praxis case therefore functions as a thought experiment that foregrounds normative concerns around exclusion,

Table 2
Comparative summary of case studies. Note: The PNS row summarises normative critiques derived from Wood (2025) rather than empirical observations, reflecting its role as a theoretical counterpoint in our analysis.

Dimension	Hangzhou City Brain (HCBI)	Shanghai AI development zone	Praxis Network State (PNS)
Technology	AIoT platform; traffic → multi-domain services; predictive analytics	Citywide data platform; computer vision pilots; AI labs/testbeds	Blockchain/AI governance stack; DAO/smart-contract tooling
Governance	State-led P3; municipal command "digital cockpit"; KPI-driven ops	Municipal–corporate collaboration; sandbox regulation	Private sovereignty/charter-city logic; minimal public oversight
Power	Municipal + Alibaba data integration; strong state steering	City/national strategy + big-tech alliances; policy entrepreneurs	Venture-capital dominance; investor–founder control
Spatiality	Embedded in existing fabric; citywide integration of services	Integrated into transport/logistics corridors; campus/test districts	New enclave; access-controlled urban forms; perimeterised spaces
Autonomy	Mixed human/AI decision-making; escalating automation in operations	Algorithmic recommendations → selective auto-execution pilots	Algorithmic rule enforcement; automated governance
Ethics	Health-code privacy issues; consent ambiguity; equity concerns	Surveillance/facial-recognition debates; transparency controversies	Exclusionary citizenship; rights deficits; eugenics critiques
Economy	Efficiency gains; procurement dependence; limited public data value-share	Innovation incentives; platformisation of services; startup clustering	Hyper-capitalist/crypto-finance; tokenised land/property
Culture	Convenience-led acceptance; "smart Hangzhou" branding	Innovation showcase/expo culture; citizen super-apps	Elite techno-utopian identity; membership-based community
Materiality	Dense CCTV/sensors; edge devices; control rooms; some service robots	Extensive IoT/CCTV; data centres; AV/drone testbeds	Purpose-built enclave; heavy surveillance hardware; drones/checkpoints

citizenship and algorithmic sovereignty rather than as an empirical field study. Owing to the emergent status of urban AI and the absence of publicly available primary data in some contexts, our empirical analysis relies on secondary sources—including published case studies, policy documents, interviews reported in the literature, industry white papers and media reports—to construct our case narratives. We adopt a secondary-data synthesis approach, cross-checking multiple sources to establish factual consistency and analytic plausibility. This methodology aligns with the exploratory objectives of this article: to map how the nine dimensions manifest across heterogeneous contexts and to derive governance and ethical implications. We recognise that reliance on secondary data limits the granularity of our findings; the cases should thus be interpreted as illustrative rather than definitive. We encourage future research using ethnography, in-depth interviews and participatory methods to enrich and contextualise the patterns identified here.

4. Findings & discussion

Because our empirical evidence derives from secondary sources, the findings presented here are drawn from narrative reviews of published case studies, policy documents and scholarly analyses. Each case summary applies the nine analytical dimensions to existing evidence rather than reporting new fieldwork. Accordingly, the discussion emphasises cross-case patterns and theoretical implications rather than original empirical results.

4.1. Synthesis of findings (Narrative Review)

Across the three case studies, several commonalities emerge. All rely on AIoT technologies that collect data and act autonomously, signalling a shift from sensor-based smart urbanism to AI-driven post-smart systems. Each case features a convergence of state and corporate interests—from municipal–platform alliances in Hangzhou and Shanghai to investor-led corporate sovereignty in network-state proposals. In all settings, AI is promoted as a tool for efficiency—optimising traffic, logistics or service delivery—yet each simultaneously raises concerns about surveillance and social control, whether through health-code systems, predictive policing or exclusionary citizenship models. Important divergences also surface. Hangzhou and Shanghai represent state-led experiments embedded in national strategies and public–private collaboration; citizen participation is limited but present through service delivery and feedback channels. By contrast, the Praxis Network State epitomises a corporate-driven model with minimal public accountability. The Chinese cases retrofit AI into existing urban fabrics, while the network-state vision proposes entirely new enclaves with alternative regulatory regimes and social contracts, including transport and mobility transformations with wide social ramifications (Bissell et al., 2020). These cases illustrate how AI urbanism diverges from smart urbanism across the nine dimensions: technology (AIoT vs IoT), governance (algorithmic vs human-mediated decisions), power (state-corporate alliances vs private sovereignty), spatiality (visible hardware and new urban forms), autonomy (automation vs autonomous AI systems), ethics (bias, accountability, surveillance), economy (platform capitalism and charter cities), culture (anthropomorphised AI and everyday AI culture), and materiality (heavy robotics, AV fleets, control rooms and data centres).

4.2. Governance and socio-technical synergy

AI reconfigures governance by shifting from technology-as-tool to AI-as-stakeholder. In Hangzhou and Shanghai, “City Brain” architectures and predictive platforms increasingly steer operational choices, moving along a continuum from human-in-the-loop to machine-in-the-loop governance (governance/autonomy). This reconfiguration is inseparable from power (who sets objectives, owns data, controls platforms), ethics (accountability for algorithmic error or bias), economy (procurement regimes, vendor lock-in, platform monetisation), spatiality/materiality (where and how AI acts in public space and what infrastructures it requires), and culture (public trust, legitimacy and AI literacy). In parallel, AI localism shows cities are not waiting for national governments: municipal charters and ordinances are setting principles, transparency requirements, engagement procedures and responsible procurement rules for AI systems (Marcucci et al., 2022). Governance must also adapt to agentic AI, where systems can set sub-goals and modify behaviour, raising novel oversight and escalation challenges (Tiwari, 2025). Furthermore, systematic reviews highlight the growing coupling of AI with digital twins, shifting AI from analytical support toward co-decision-making across planning workflows (Lartey & Law, 2025). Achieving socio-technical synergy therefore requires: (a) multi-stakeholder design and oversight that redistribute power (e.g., citizen assemblies for high-impact deployments); (b) governance processes that mandate explainability and contestability (e.g., public algorithm registers with appeal channels); (c) open standards and public digital infrastructure to mitigate vendor lock-in (e.g., open APIs and data-portability clauses in procurement contracts); and (d) spatial rules for robots/AVs that protect public-space rights (e.g., geofenced slow-zones and accessibility-first curb codes). Without these guardrails, AI risks entrenching technocratic decision-making and eroding legitimacy.

4.3. Power, autonomy and human–machine relations

AI intensifies power asymmetries by concentrating control of data, compute and strategic assets within state agencies and large platforms (power/economy). This dynamic resonates with surveillance capitalism, in which data extraction by platform actors threatens democratic accountability (Makanadar, 2024). As systems gain autonomy, decision prerogatives shift from human operators to machine agents (autonomy/governance), reshaping human–machine relations in frontline domains (mobility, safety, welfare). Generative AI additionally enables anticipatory governance, where model-produced futures steer present action—risking over-reliance on “AI oracles” (Cugurullo & Xu, 2025). These trends intersect with algorithmic governmentality, which can translate optimisation logics into spatialised inequities and forms of spatial injustice (Rodrigues, 2016). Protecting human agency requires clear allocation of authority (who is ultimately accountable), fail-safes and escalation protocols (governance/autonomy), rights to explanation and appeal (ethics), and institutional capacity for algorithmic auditing (power/ethics), complemented by AI literacy and co-design practices (culture/governance).

4.4. Ethical, social and cultural implications

Urban planning applications of AI face persistent risks around bias, transparency, accountability and privacy, calling for inclusive datasets, meaningful public engagement and continuous monitoring (Sanchez et al., 2025). Municipalities should institute algorithmic-accountability and transparency practices—documenting objectives, datasets, training/validation protocols, impact assessments and audit logs—and narrow trade-secret shields that impede public scrutiny (Baykurt, 2022; Brauneis & Goodman, 2018). Legal analyses of AIoT deployments likewise foreground data-protection, liability and human-rights safeguards as baseline requirements (Rawindaran, 2023; Shafik, 2024). Environmentally, AI's footprint—energy use, water consumption, CO_2 emissions and e-waste—demands lifecycle governance aligned with sustainability goals (Palmini & Cugurullo, 2024). On the material front, robotics and automation extend or displace human agency and can intensify socio-spatial sorting if not governed responsibly (Macrorie et al., 2021). Driverless vehicles carry broad social impacts beyond safety and throughput, including labour transformations and new inequalities (Bissell et al., 2020). Culturally, everyday interactions with anthropomorphised agents (named robots, voice assistants) normalise AI as a social actor, potentially blurring lines of accountability (culture/power/ethics).

4.5. Political economy: Social control and hyper-capitalist development

Urban AI underwrites dual political-economic logics. On one hand, it extends social control through predictive governance and ambient monitoring (power/governance). On the other, platform ecosystems capture and monetise urban big data, consolidating infrastructural power and creating new modes of accumulation (Barns, 2020). These logics often co-produce each other: tighter control facilitates accumulation; accumulation finances more control. These dynamics are not abstract: as Safransky (2020) shows, algorithmic systems can produce "geographies of algorithmic violence," reinscribing redlining-style discrimination across credit, policing, housing allocation and access to services. An anti-oppressive policy design for urban AI therefore requires equity impact assessments, redistributive data governance and community oversight to prevent the codification of historical harms (Safransky, 2020). Postcolonial critiques show how smart/AI city imaginaries can reproduce neoliberal and colonial narratives that marginalise alternative futures (Datta, 2019; Cardullo & Kitchin, 2019). At the same time, decentralised models—including data cooperatives and community-owned infrastructures—offer pathways that challenge surveillance-capitalist dynamics and recentralise value to communities (Calzada, 2024). The resulting tensions foreground municipal sovereignty, public value capture and the design of countervailing civic institutions (power/economy).

4.6. Spatiality and materiality

Post-smart urbanism remakes space and matter. The shift from invisible sensors to visible robots, drones and AVs requires new spatial codes (curb management, right-of-way, geofencing) and material infrastructures (charging, maintenance depots, control rooms, edge compute). Policy design for robots in public spaces must therefore encompass safety-by-design, privacy safeguards and pathways to public acceptance and legitimacy (Mintrom et al., 2022). Urban robotics research shows how experiments can reproduce socio-spatial sorting and must therefore be governed through robust public interest tests (Macrorie et al., 2021). Empirically, urban experiments in San Francisco, Tokyo and Dubai demonstrate how automation pilots are territorially staged and institutionally mediated to test governance and public acceptance (While et al., 2021). Parallel developments in controlled sites such as airports show how service and security robots recalibrate everyday routines, labour and spatial practices—often prefiguring citywide adoption (Lin & Yeo, 2023). Algorithmic planning also indirectly reshapes land use and flows, privileging optimised corridors and logistics hubs (spatiality/technology). At the same time, as Mattern (2020) reminds us, a city is not a computer: analogue infrastructures, tacit knowledge and care work should remain foregrounded so that computational systems augment—rather than overwrite—human judgement and situated practices. To safeguard publicness and equity, cities need design standards for human–robot interaction zones, inclusive accessibility rules and environmental limits on fleets and data-centre growth (spatiality/materiality/ethics).

4.7. Toward a normative framework for post-smart urbanism

A normative framework for post-smart urbanism must align technical capabilities with public value across the nine analytical dimensions—technology, governance, power, spatiality, autonomy, ethics, economy, culture and materiality—so that the adoption of AI advances democratic ends rather than undermining them. At its core, governance is where these dimensions are reconciled: the assignment of decision rights, oversight arrangements and accountability mechanisms determines whether algorithmic systems remain instruments of public purpose or drift toward opaque infrastructural power. This implies redistributing power through public and civic counterweights to platform dominance; instituting robust procedures for explanation and contestation; and ensuring that public authorities, not vendors, set goals and norms. Because autonomy in machine systems can displace human agency, the framework mandates "meaningful human control," with escalation thresholds, fail-safe modes and ultimate accountability vested in identifiable public actors.

Ethical commitments—fairness, privacy, transparency and the right to redress—must be embedded across the lifecycle, supported by inclusive datasets, public engagement and human oversight (Sanchez et al., 2025). Building on ethical frameworks for AI and sustainable cities, this further entails explicit commitments to fairness, human well-being and ecological limits that connect model design and deployment to measurable urban sustainability outcomes (Pastor-Escuredo et al., 2022). Economic arrangements should avoid lock-in via interoperability and open standards, complemented by AI-localism lessons on principles, accountability and responsible procurement (Marcucci et al., 2022). Spatiality and materiality require design codes for human–robot interaction, curb/sidewalk governance and environmental performance standards for data centres, edge compute and fleets (Palmini & Cugurullo, 2024; Macrorie et al., 2021). The framework should also enable decentralised governance via data cooperatives and civic cloud to advance data sovereignty—operationalised as community-empowering models that redistribute data value, decision rights and accountability to residents and local institutions (Calzada, 2024)—and incorporate guidance for anticipatory governance in the era of generative AI (Cugurullo & Xu, 2025). Cultural legitimacy must be cultivated through AI literacy, co-design with affected communities and transparency intelligible to non-experts. Together, these commitments create an integrated architecture in which innovation is a means to collectively defined social purposes.

4.8. Policy recommendations and future research

Operationalising this framework calls for coordinated action at multiple scales. Cities and higher tiers of government should establish multi-level regimes that clarify authority over algorithmic systems, specify duties of care and create independent oversight with powers to audit, sanction and remediate. Legal obligations for transparency and explainability should extend to rights of appeal and accessible channels for contestation by individuals and civil society. Public investment in shared digital infrastructure—open APIs, trustworthy identity, secure civic cloud and sandboxing—can counterbalance corporate dominance, while procurement should prioritise interoperability, portability and explicit exit strategies, reflecting AI-localism practice (Marcucci et al., 2022). Capacity building is essential: organisational readiness, change management and evaluation processes are critical for generative-AI adoption (Chandy, 2025). Strategic roadmaps should align with urban AI frameworks that integrate technical systems, applications, education and governance (Zhu & Liu, 2025). Environmental due diligence—energy intensity, water use, embodied carbon, end-of-life recovery—should be mandatory for major deployments, with binding targets tied to climate and circular-economy commitments (Palmini & Cugurullo, 2024).

Beyond these near-term actions, future research should explicitly incorporate posthumanist perspectives on intelligence, autonomy and agency, and deploy narrative/semantic-network analyses to trace how AI imaginaries materialise in policy and space (Iapaolo & Lynch, 2025; Lazzeroni & Romano, 2025). Future research should (i) conduct longitudinal studies on equity, wellbeing and institutional legitimacy; (ii) pursue comparative work across the global North/South to avoid universalising narrow cases; (iii) experimentally evaluate alternative institutional designs such as community-owned AI services, public–civic data trusts, cooperative cloud and open-source platforms; (iv) deepen theoretical lenses via posthumanist analyses of intelligence, autonomy and agency (Iapaolo & Lynch, 2025); and (v) analyse the urban narratives and imaginaries that AI engenders using semantic/network approaches (Lazzeroni & Romano, 2025). An explicitly interdisciplinary agenda spanning urban studies, computer science, law, ethics and political theory remains essential to address the entanglements of technology, governance, power, spatiality, autonomy, culture, economy and materiality that define the post-smart city.

5. Conclusion

This study has argued that the advent of Urban AI marks a break with the sensor-centric smart city, ushering cities into a post-smart condition characterised by agentic systems, anticipatory governance and new political economies. Through a comprehensive literature synthesis and comparative case analysis, we demonstrated that AI urbanism is a distinct socio-technical formation with profound implications for governance, power, spatiality, autonomy, ethics, economy, culture and materiality. The nine-dimensional framework developed here provides a conceptual toolkit for analysing post-smart cities, revealing how AI reshapes public space, redistributes authority and engenders new modes of surveillance and accumulation. Our case studies—from Hangzhou and Shanghai to the Praxis Network State—illustrate that AI urbanism emerges through diverse political economies, blending state ambitions, corporate interests and technological imaginaries. These experiments promise efficiency and innovation but reveal systemic risks: concentration of data and compute power, creeping algorithmic governance, erosion of citizen agency and deepening socio-spatial inequalities.

Beyond delineating the shift from sensor-centric smart cities to AI-enabled agentic urbanism, this article makes three distinct contributions. First, it offers an integrative synthesis of scholarship on smart cities, AI urbanism, platform governance, innovation ecosystems and digital authoritarianism, weaving insights from urban studies, planning, ethics and innovation into nine interrelated dimensions—technology, governance, power, spatiality, autonomy, ethics, economy, culture and materiality. This synthesis clarifies conceptual boundaries and demonstrates how AI urbanism constitutes a qualitatively different socio-technical regime. Second, it proposes a conceptual toolkit for analysing post-smart cities: the nine-dimensional framework and a socio-technical model that link AI capabilities, governance mechanisms and socio-technical ecosystems. Applied to narrative case reviews of Hangzhou, Shanghai and the Praxis Network State, the toolkit enables systematic comparison across contexts and supports evaluative judgements. Third, the paper bridges conceptual analysis with normative guidance by articulating a human-centred governance framework that emphasises inclusive decision-making, transparency, data sovereignty and environmental stewardship, offering practitioners a roadmap for aligning AI innovation with public value. By integrating theoretical synthesis, methodological innovation and normative policy guidance, this study advances both scholarly and practical understandings of post-smart urbanism.

The analysis underscores that governance is the pivot at which technical, ethical and political dimensions converge. Absent robust accountability mechanisms, public oversight and participatory design, AI may reinforce technocracy and authoritarian tendencies. Conversely, AI localism and emerging community-centred models signal that alternative trajectories are possible if cities assert autonomy, establish public-interest guardrails and cultivate AI literacy. The normative framework proposed here calls for redistributing data value through civic data trusts and cooperatives; mandating transparency, explainability and contestability; incorporating environmental due diligence and Green AI principles; and designing spatial regulations for embodied AI systems. These measures can enable socio-technical synergies that align AI deployment with democratic values, social equity and ecological sustainability.

Several limitations deserve acknowledgement. Our reliance on secondary sources constrains the granularity of case insights and may miss grassroots perspectives. The cases—while diverse—do not capture the full spectrum of global AI urbanism, particularly in the Global South where resource constraints and governance structures differ. Future research should therefore engage in ethnographic studies of AI implementation, longitudinal analysis of social impacts and comparative work across varied contexts. Interdisciplinary collaborations among urbanists, technologists, ethicists and legal scholars are essential to refine theoretical models and develop empirically grounded policy interventions. Ultimately, the post-smart urbanism is not an inevitable, technocratic destiny but a contested terrain. By embracing participatory governance, ethical design and ecological responsibility, scholars and practitioners can shape AI urbanism toward inclusive and sustainable urban futures.

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Declaration of competing interest

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Appendix A. Supplementary data

Supplementary Table S1 provides a structured reference map linking key sources to the nine analytical dimensions of post-smart urbanism, supporting transparency and traceability of the synthesis presented in Section 2. Supplementary data to this article can be found online at https://doi.org/10.1016/j.tele.2026.102399.

Data availability

No data was used for the research described in the article.

References

Allam, Z., Dhunny, Z.A., 2019. On big data, artificial intelligence and smart cities. Cities 89, 80–91. https://doi.org/10.1016/j.cities.2019.01.032.

Bai, O., Shi, Y., Wu, F., 2025. Pioneering the post-smart cities paradigm in urban innovation ecosystems governance: a case study of hangzhou city brain initiative. Urban Policy Res. 1–25. https://doi.org/10.1080/08111146.2025.2456258.

Barns, S., 2020. Re-engineering the city: platform ecosystems and the capture of urban big data. Front. Sustainable Cities 2, 32. https://doi.org/10.3389/frsc.2020.00032.

Baykurt, B., 2022. Algorithmic accountability in US cities: transparency, impact, and political economy. Big Data Soc. 9 (2), 20539517221115426. https://doi.org/10.1177/20539517221115426.

Bissell, D., Birtchnell, T., Elliott, A., Hsu, E.L., 2020. Autonomous automobilities: the social impacts of driverless vehicles. Curr. Sociol. 68 (1), 116–134. https://doi.org/10.1177/0011392118816743.

Bosco, G., Riccardi, V., Sciarrone, A., D'Amore, R., Visvizi, A., 2024. AI-driven innovation in smart city governance: achieving human-centric and sustainable outcomes. Transforming Government: People, Process and Policy. https://doi.org/10.1108/TG-04-2024-0096.

Bratton, B., 2021. AI urbanism: a design framework for governance, program, and platform cognition. AI & Soc. 36 (4), 1307–1312. https://doi.org/10.1007/s00146-020-01121-9.

Brauneis, R., Goodman, E.P., 2018. Algorithmic transparency for the smart city. Yale JL & Tech. 20, 103–139. https://doi.org/10.2139/ssrn.3012499.

Calzada, I., 2024. Artificial intelligence for social innovation: beyond the noise of algorithms and datafication. Sustainability 16 (19), 8638. https://doi.org/10.3390/su16198638.

Caprotti, F., Chang, I.C.C., Joss, S., 2022. Beyond the smart city: a typology of platform urbanism. Urban Transform. 4 (1), 4. https://doi.org/10.1186/s42854-022-00033-9.

Caprotti, F., Cugurullo, F., Cook, M., Karvonen, A., Marvin, S., McGuirk, P., Valdez, A.M., 2024. Why does urban Artificial Intelligence (AI) matter for urban studies? developing research directions in urban AI research. Urban Geogr. 45 (5), 883–894. https://doi.org/10.1080/02723638.2024.2329401.

Cardullo, P., Kitchin, R., 2019. Smart urbanism and smart citizenship: the neoliberal logic of 'citizen-focused'smart cities in Europe. Environment and Planning c: Politics and Space 37 (5), 813–830. https://doi.org/10.1177/0263774X18806508.

Chandy, K. (2025). *AI on the Street: An Exploration of Generative AI in City Ecosystems* (Master's thesis, Harvard University). Available at: https://dash.harvard.edu/handle/1/42719875 (accessed 15 July. 2025).

Chaturvedi, V., de Vries, W.T., 2021. Machine learning algorithms for urban land use planning: a review. Urban Sci. 5 (3), 68. https://doi.org/10.3390/urbansci5030068.

Cugurullo, F., 2020. Urban artificial intelligence: from automation to autonomy in the smart city. Front. Sustainable Cities 2, 38. https://doi.org/10.3389/frsc.2020.00038.

Cugurullo, F., 2021. "One AI to rule them all": the unification of chinese urban governance under artificial intelligence. GREEN 1 (1), 123–126. https://doi.org/10.3917/green.001.0134.
Cugurullo, F., Xu, Y., 2025. When AIs become oracles: generative artificial intelligence, anticipatory urban governance, and the future of cities. Policy and Society 44 (1), 98–115. https://doi.org/10.1093/polsoc/puae025.
Cugurullo, F., Caprotti, F., Cook, M., Karvonen, A., McGuirk, P., Marvin, S., 2024. The rise of AI urbanism in post-smart cities: a critical commentary on urban artificial intelligence. Urban Stud. 61 (6), 1168–1182. https://doi.org/10.1177/00420980231203386.
Cugurullo, F., Caprotti, F., Cook, M., Karvonen, A., McGuirk, P., & Marvin, S. (Eds.). (2023). *Artificial intelligence and the city: Urbanistic perspectives on AI*. Taylor & Francis. https://doi.org/10.4324/9781003365877.
Datta, A., 2019. Postcolonial urban futures: imagining and governing India's smart urban age. Environ. Plann. D: Soc. Space 37 (3), 393–410. https://doi.org/10.1177/0263775818800721.
Datta, A., Odendaal, N., 2019. Smart cities and the banality of power. Environ. Plann. D: Soc. Space 37 (3), 387–392. https://doi.org/10.1177/0263775819841765.
Ejjami, R., 2024. AI-driven smart cities in France. International Journal for Multidisciplinary Research 6 (3), 37–51. https://doi.org/10.36948/ijfmr.2024.v06i03.21920.
Guo, Z., Cugurullo, F., 2025. Smart urbanism through artificial intelligence (AI)-megaprojects: the case of China's healthcare services. Public Adm. Dev. 45 (1). https://doi.org/10.1002/pad.2111.
He, W., Chen, M., 2024. Advancing urban life: a systematic review of emerging technologies and artificial intelligence in urban design and planning. Buildings 14 (3), 835. https://doi.org/10.3390/buildings14030835.
Iapaolo, F., Lynch, C.R., 2025. Rethinking urban AI: a critical posthumanist perspective on intelligence, autonomy, and agency. Urban Geogr. 1–23. https://doi.org/10.1080/02723638.2025.2510812.
Joss, S., Sengers, F., Schraven, D., Caprotti, F., Dayot, Y., 2019. The smart city as global discourse: storylines and critical junctures across 27 cities. J. Urban Technol. 26 (1), 3–34. https://doi.org/10.1080/10630732.2018.1558387.
Katzenbach, C., Ulbricht, L., 2019. Algorithmic governance. Internet Policy Review 8 (4), 1–18. https://doi.org/10.14763/2019.4.1424.
Kitchin, R., Coletta, C., Evans, L., Heaphy, L., Mac Donncha, D., 2019. Smart cities, algorithmic technocracy and new urban technocrats. In: Planning and Knowledge. Policy Press, pp. 199–212.
Krivý, M., 2018. Towards a critique of cybernetic urbanism: the smart city and the society of control. Plan. Theory 17 (1), 8–30. https://doi.org/10.1177/1473095216645631.
Kummitha, R.K.R., Crutzen, N., 2019. Smart cities and the citizen-driven internet of things: a qualitative inquiry into an emerging smart city. Technol. Forecast. Soc. Chang. 140, 44–53. https://doi.org/10.1016/j.techfore.2018.12.001.
Lartey, D., Law, K.M., 2025. Artificial intelligence adoption in urban planning governance: a systematic review of advancements in decision-making, and policy making. Landsc. Urban Plan. 258, 105337. https://doi.org/10.1016/j.landurbplan.2025.105337.
Lazzeroni, M. (2023). Artificial intelligence and new visions of the City: Assemblages, Imaginaries, Policies. In *Geography of our common future*. https://arpi.unipi.it/handle/11568/1216149.
Lazzeroni, M., Romano, A., 2025. Artificial intelligence and new visions of the future of the city: exploring urban narratives through semantic and network analysis. J. Urban Technol. 32 (2), 63–83. https://doi.org/10.1080/10630732.2025.2469326.
Leszczynski, A., 2016. Speculative futures: cities, data, and governance beyond smart urbanism. Environ. Plann. A: Econ. Space 48 (9), 1691–1708. https://doi.org/10.1177/0308518X16651445.
Leszczynski, A., 2020. Glitchy vignettes of platform urbanism. Environ. Plann. D: Soc. Space 38 (2), 189–208. https://doi.org/10.1177/0263775819878721.
Lin, W., Yeo, S.J.I., 2023. Airport robots: automation, everyday life and the futures of urbanism. In: Artificial Intelligence and the City. Routledge, pp. 169–185. https://doi.org/10.4324/9781003365877-13.
Luque-Ayala, A., Marvin, S., 2020. *Urban operating systems: producing the computational city*. MIT Press. https://doi.org/10.7551/mitpress/10869.001.0001.
Luusua, A., Ylipulli, J., Foth, M., Aurigi, A., 2023. Urban AI: Understanding the emerging role of artificial intelligence in smart cities. AI & Soc. 38 (3), 1039–1044. https://doi.org/10.1007/s00146-022-01537-5.
Macrorie, R., Marvin, S., While, A., 2021. Robotics and automation in the city: a research agenda. Urban Geogr. 42 (2), 197–217. https://doi.org/10.1080/02723638.2019.1698868.
Makanadar, A., 2024. Digital surveillance capitalism and cities: data, democracy and activism. Humanities and Social Sciences Communications 11 (1), 1–7. https://doi.org/10.1057/s41599-024-03941-2.
Marcucci, S., Kalkar, U., & Verhulst, S. (2022). AI localism in practice: Examining how cities govern AI. *Available at SSRN 4284013*. https://doi.org/10.2139/ssrn.4284013.
Marvin, S., While, A., Chen, B., Kovacic, M., 2022. Urban AI in China: Social control or hyper-capitalist development in the post-smart city? Front. Sustainable Cities 4, 1030318. https://doi.org/10.3389/frsc.2022.1030318.
Mattern, S., 2020. A city is not a computer. In: *The Routledge Companion to Smart Cities*. Routledge, pp. 17–28. https://doi.org/10.4324/9781315178387-2.
Mintrom, M., Sumartojo, S., Kulić, D., Tian, L., Carreno-Medrano, P., Allen, A., 2022. Robots in public spaces: implications for policy design. Policy Design and Practice 5 (2), 123–139. https://doi.org/10.1080/25741292.2021.1905342.
Ortega-Fernández, A., Martín-Rojas, R., García-Morales, V.J., 2020. Artificial intelligence in the urban environment: Smart cities as models for developing innovation and sustainability. Sustainability 12 (19), 7860. https://doi.org/10.3390/su12197860.
Palmini, O., Cugurullo, F., 2023. Charting AI urbanism: conceptual sources and spatial implications of urban artificial intelligence. Discover Artificial Intelligence 3 (1), 15. https://doi.org/10.1007/s44163-023-00060-w.
Palmini, O., Cugurullo, F., 2024. Design culture for Sustainable urban artificial intelligence: Bruno Latour and the search for a different AI urbanism. Ethics Inf. Technol. 26 (1), 11. https://doi.org/10.1007/s10676-024-09752-y.
Pastor-Escuredo, D., Treleaven, P., Vinuesa, R., 2022. An ethical framework for artificial intelligence and sustainable cities. Ai 3 (4), 961–974. https://doi.org/10.3390/ai3040057.
Rawindaran, N., 2023. Legal considerations and ethical challenges of artificial intelligence on internet of things and smart cities. In: *Data Protection in a Post-Pandemic Society: Laws, Regulations, Best Practices and Recent Solutions*. Springer International Publishing, Cham, pp. 217–239. https://doi.org/10.1007/978-3-031-34006-2_8.
Rodrigues, N., 2016. Algorithmic governmentality, smart cities and spatial justice. Justice Spatiale= Spatial Justice 10.
Sadowski, J., Bendor, R., 2019. Selling smartness: corporate narratives and the smart city as a sociotechnical imaginary. Sci. Technol. Hum. Values 44 (3), 540–563. https://doi.org/10.1177/0162243918806061.
Safransky, S., 2020. Geographies of algorithmic violence: Redlining the smart city. Int. J. Urban Reg. Res. 44 (2), 200–218. https://doi.org/10.1111/1468-2427.12833.
Sanchez, T.W., Brenman, M., Ye, X., 2025. The ethical concerns of artificial intelligence in urban planning. J. Am. Plann. Assoc. 91 (2), 294–307. https://doi.org/10.1080/01944363.2024.2355305.
Shafik, W., 2024. Ethical use of machine learning techniques in smart cities. In: *Ethical Artificial Intelligence in Power Electronics*. CRC Press, pp. 21–47. https://doi.org/10.1201/9781032648323-3.
Tiwari, A., 2025. Conceptualising the emergence of Agentic Urban AI: from automation to agency. Urban Inf. 4 (1), 13. https://doi.org/10.1007/s44212-025-00079-7.
Verrest, H., Pfeffer, K., 2019. Elaborating the urbanism in smart urbanism: distilling relevant dimensions for a comprehensive analysis of Smart City approaches. Inf. Commun. Soc. 22 (9), 1328–1342. https://doi.org/10.1080/1369118X.2018.1424921.
While, A.H., Marvin, S., Kovacic, M., 2021. Urban robotic experimentation: San Francisco. Tokyo and Dubai. Urban Studies 58 (4), 769–786. https://doi.org/10.1177/0042098020917790.

Wood, D.M., 2025. Post-smart cities as digital authoritarian polities: the aesthetic politics of the Praxis Network State. Dialogues on Digital Society. https://doi.org/10.1177/29768640251377168.

Yigitcanlar, T., Mehmood, R., Corchado, J.M., 2021. Green artificial intelligence: towards an efficient, sustainable and equitable technology for smart cities and futures. Sustainability 13 (16), 8952. https://doi.org/10.3390/su13168952.

Zhang, J., Hua, X.S., Huang, J., Shen, X., Chen, J., Zhou, Q., Zhao, Y., 2019. City brain: practice of large-scale artificial intelligence in the real world. IET Smart Cities 1 (1), 28–37. https://doi.org/10.1049/iet-smc.2019.0034.

Zhu, D., Liu, H., 2025. City AI: a strategic framework for urban artificial intelligence application and development. Urban Inf. 4 (1), 12. https://doi.org/10.1007/s44212-025-00077-9.